dortmund university

Economic MPC for Room Temperature Control

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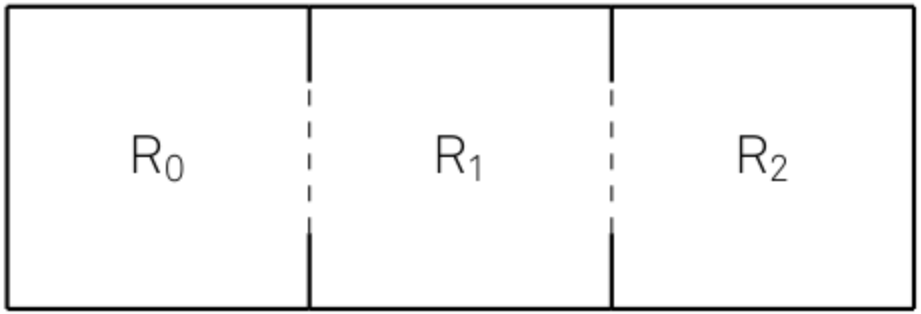
Introduction

Introduction

- Model Predictive Control
  - Enhances energy efficiency and sustainable building management.
  - Incorporates weather forecasts to adjust heating and cooling strategies preemptively.
  - Tailored to expected environmental conditions.
  - Ensures occupant comfort while minimizing energy waste.
- Kalman Filter as a state observer.

Methodology

System Model



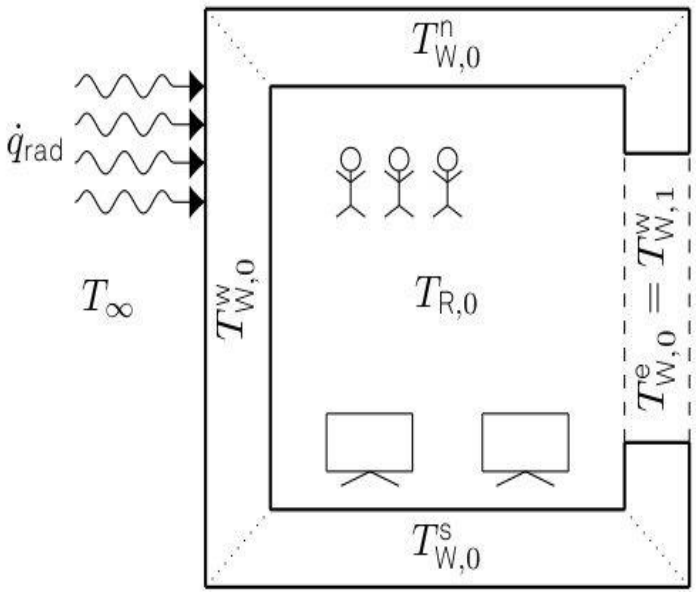
The building system model, illustrated in the figures, encompasses the interaction between room air temperature, wall temperatures, heat sources, and environmental parameters, enabling MPC to effectively control room temperature and optimize energy use.

System Model

Components of a Room

- Room Temperatures $T_{R,i}$: Temperature of the air within the room.
- Wall Temperatures $T_{W,i}^{d}$: Wall temperatures in different directions (north, east, south, west).
- Other Heat Sources: Heat produced by humans and PCs.

- Room temperature $\frac{\partial T_{R,i}}{\partial t} = f_1(u(t), x(t))$
- Outer wall temperature $\frac{\partial T_{W,i}}{\partial t} = f_2(u(t), x(t))$
- Inner wall temperature $\frac{\partial T_{W,i}}{\partial t} = f_3(u(t), x(t))$



Linearity & Stability

- After analyzing the system ODEs, it was apparent that the system is linear.
- It can be written in state space form:

$$\dot{x}(t) = Ax\,(t) + B\,u(t) + E\,d(t) + F$$

- Negative eigen values of matrix A indicate that the system is stable.

Discretization using Implicit Euler

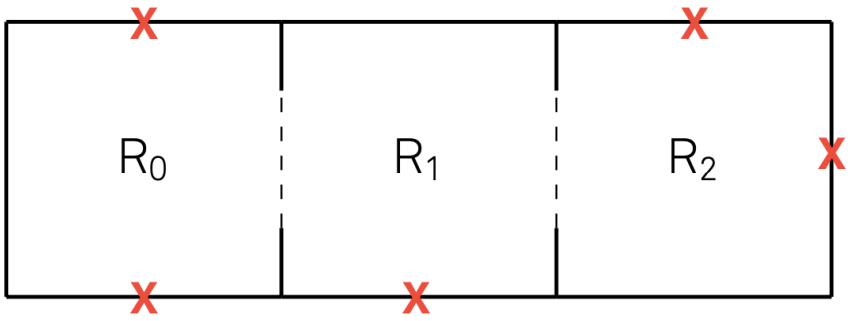
- Discretizing the model is a crucial in order to be able to employ MPC.
- Implicit Euler scheme was implemented, which is a direct full discretization approach.
- Choice of discretization step (h)

Equation:

$$x_{k+1} = x_k + h * f(x_{k+1}, u_k, d_k)$$

Kalman Filter State Observer

- It was implemented to simulate a real system behavior
- Process and measurement noise was added
- It was implemented based on the minimum number of measurements possible.
- A grid search determined that a minimum of five observers are necessary for full observability of the system.



MPC implementation

The MPC optimization problem was implemented using the CasAdi library and includes:

- Soft constraints for room temperatures using slack variables
- Pseudo-economic cost function:

$$
\begin{aligned}
l_{\text{pseudo}}(x_k, u_k, u_{k-1}) = &(x_k - x_{\text{ref}})^T Q (x_k - x_{\text{ref}}) \\
&+ (\Delta u_k)^T R (\Delta u_k) \\
&+ c^T u_k + \rho(\varepsilon_{\text{ub}}^2 - \varepsilon_{\text{lb}}^2)
\end{aligned}
$$

- Economic cost function:

$$
l_{eco}(x_k, u_k, u_{k-1}) = \Delta u_k^T R \Delta u_k + c^T u_k + \rho(\varepsilon_{ub}^2 - \varepsilon_{lb}^2)
$$

Optimization problem

$$
\begin{aligned}
\underset{u_1,\ldots,u_{N-1},x_1,\ldots,x_N,\varepsilon_{ub},\varepsilon_{lb}}{\text{minimize}} \quad & \sum_{k=0}^{N-1} l(x(k),u(k),u(k-1),\varepsilon) \\
\text{subject to} \quad & x(k+1) = x(k) + hf(x(k+1),u(k),d(k)) \\
& 19 - \varepsilon_{lb} \leq T_{R,i} \leq 23 + \varepsilon_{ub} (^\circ\mathrm{C}) \\
& 5 \leq T_{W,i} \leq 40 (^\circ\mathrm{C}) \\
& 0 \leq \dot{Q}_{heat,i} \leq 3 (kW) \\
& 0 \leq \dot{Q}_{cool,i} \leq 3 (kW) \\
& -\infty \leq \varepsilon_{ub} \leq \infty \\
& -\infty \leq \varepsilon_{lb} \leq \infty \\
& x(0) = x_{\text{init}}
\end{aligned}
$$

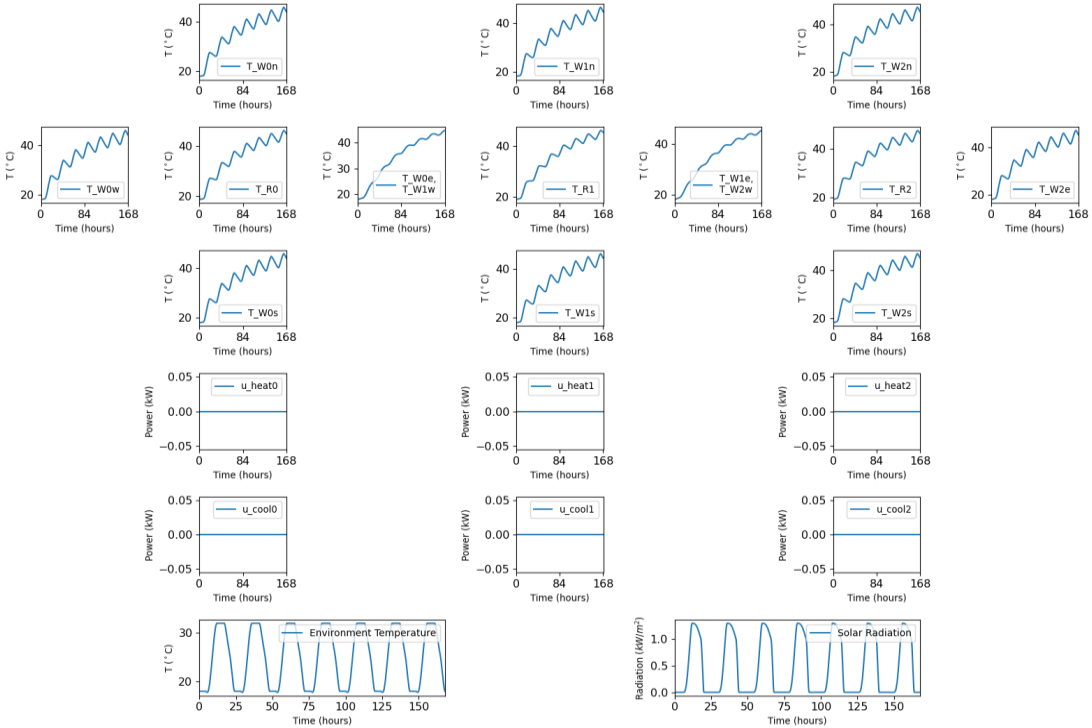
Results

Choice of h & N

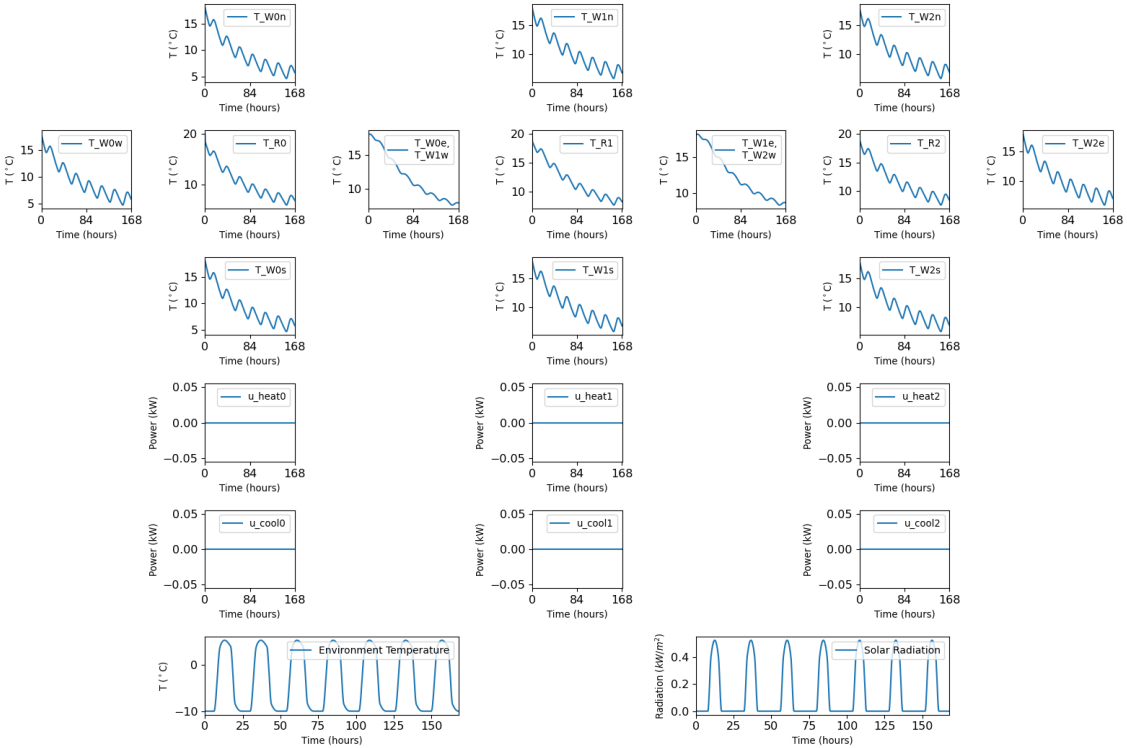
- If h is small => high complexity; if h is large => slow response and minimal adaptation
- If N is large => high complexity; if N is small => unstable MPC

- After exploring different combinations, a prediction horizon (N) of 10 hours and a discretization step (h) of 15 minutes (0.25 hours) proved to have the best balance between computational complexity, closed-loop cost and constraints violations.

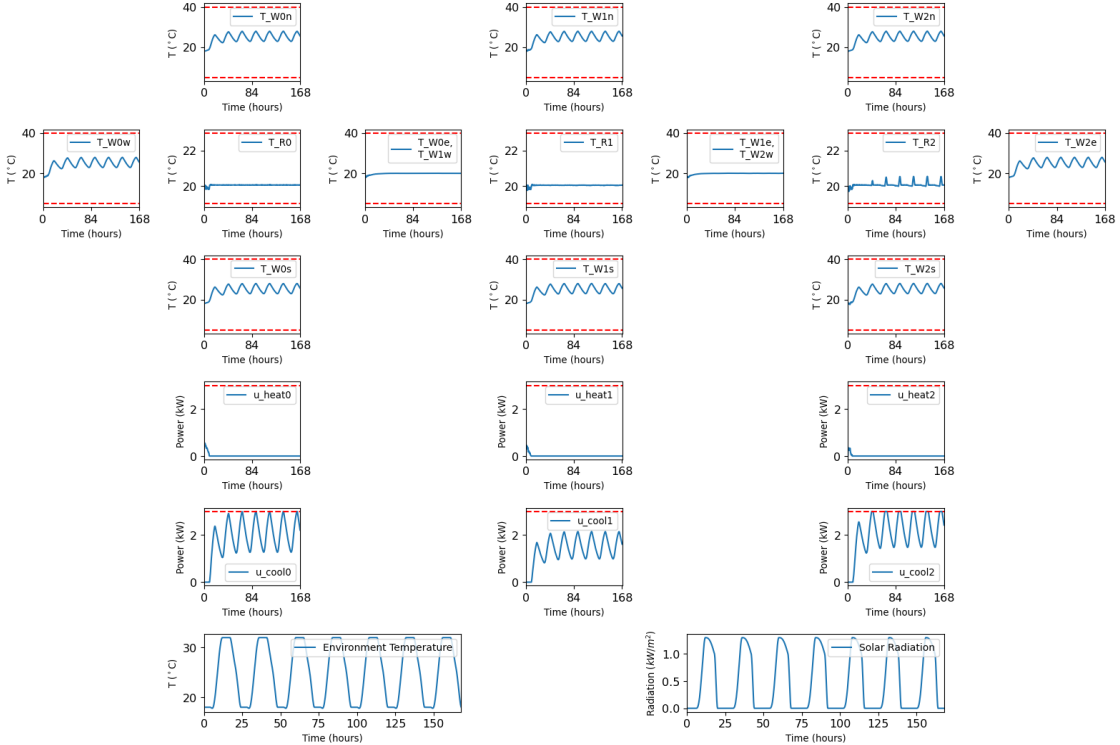
Autonomous System Behavior- Summer



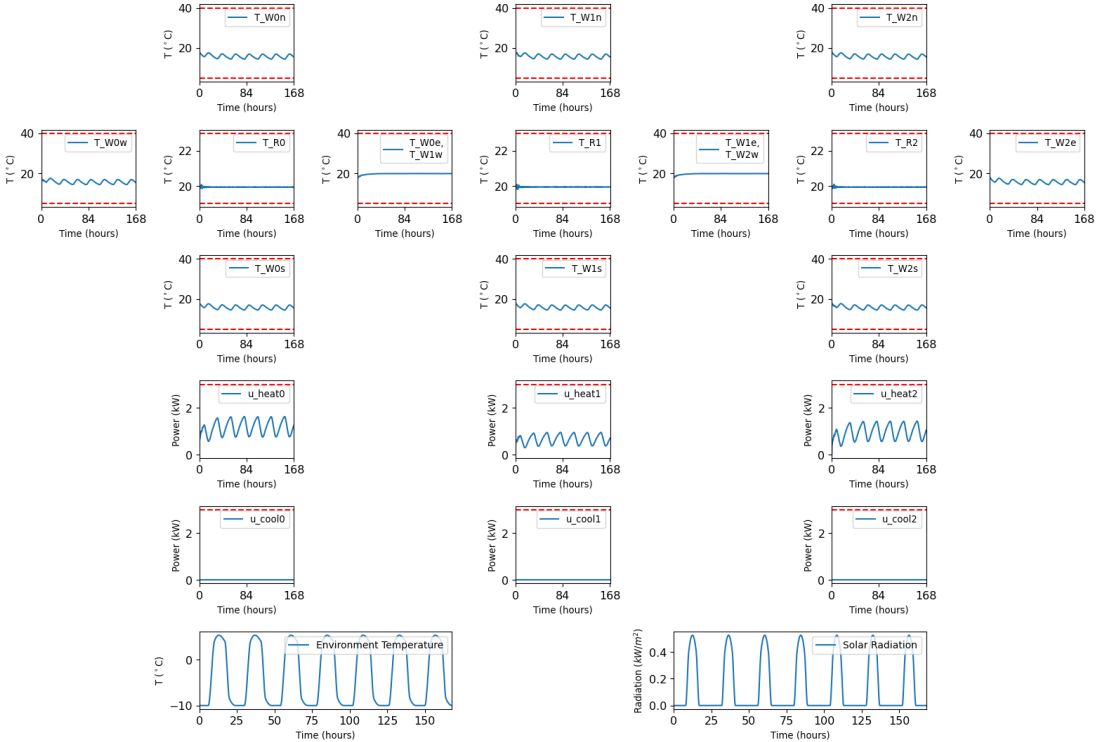
Autonomous System Behavior- Winter



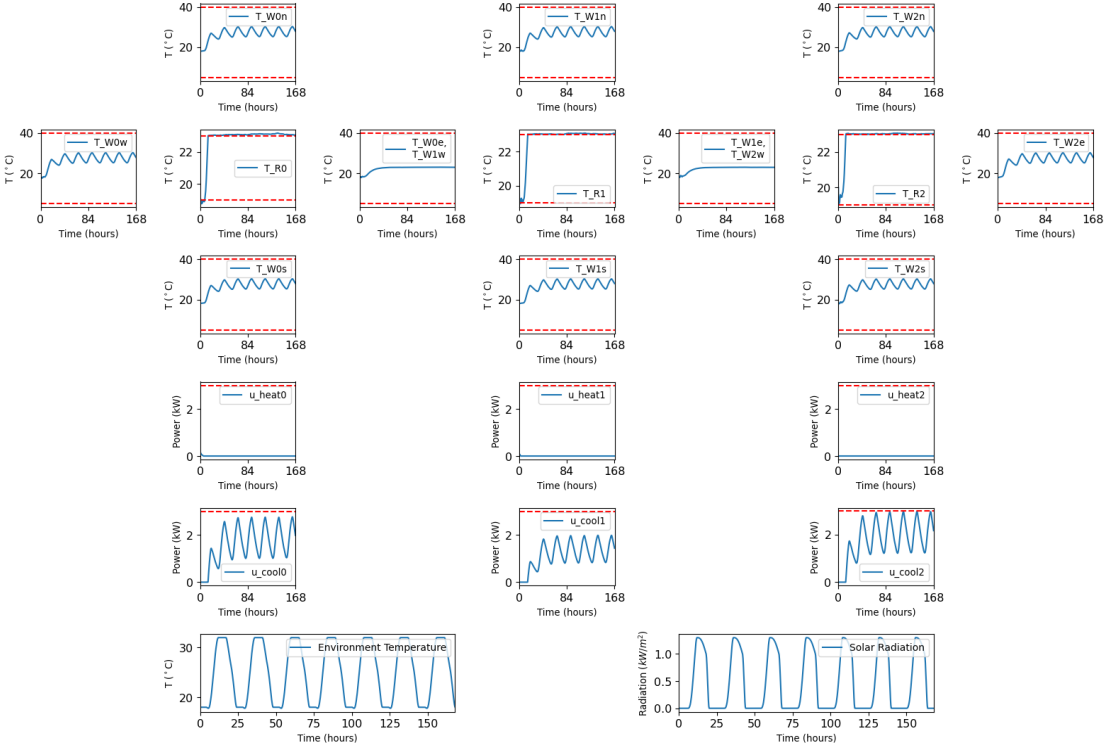
Tracking MPC - Summer



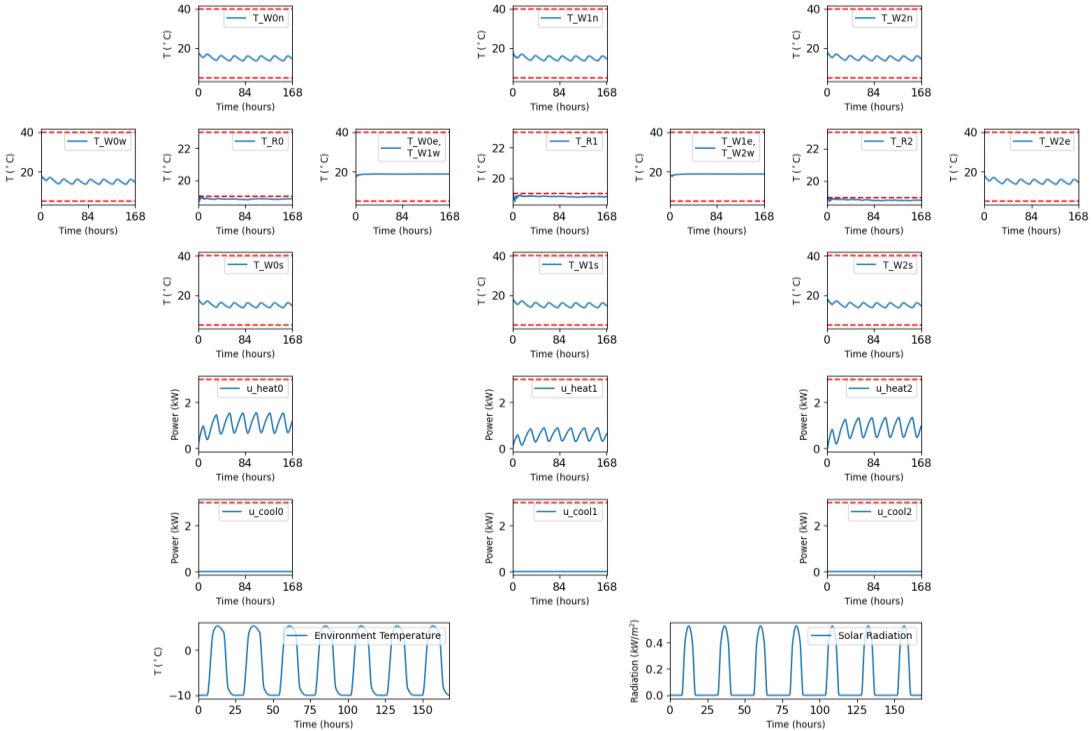
Tracking MPC - Winter



Economic MPC - Summer



Economic MPC - Winter



Performance comparison of MPCs

- **Tracking-based MPC**: Successfully stabilizes room temperatures at the reference point without constraint violations, although perfect tracking to 20°C may be impossible in summer due to cooling device limitations.
- **Economic-based MPC**: Maintains system within bounds but may experience minimal constraint violations due to the proximity to the bounds and the added noise in state estimation.
- **Closed-loop Cost**: Economic MPC generally has lower costs for input devices compared to tracking MPC, but total cost of economic MPC may be higher due to slack costs from minimal constraint violations.

Conclusion

Conclusion

- MPC proves to be the optimal controller for room temperature control to achieve sustainable building management and energy efficiency.
- Discretizing with Implicit Euler and integrating MPC with a Kalman Filter Estimator simulates real-world scenarios effectively.
- Using an economic cost function for MPC optimization is more cost-effective concerning input devices.
- Employing soft constraints, particularly in economic scenarios, ensures feasibility when constraints are slightly violated.
- Adapting weather forecasts throughout the prediction horizon greatly improves our MPC performance.